

“CDs-Cornergy” Utilization of Corn (*Zea mays L.*) Waste C-dots for Sustainable Battery Solutions

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Abstract

Lithium battery is one type of battery that is widely used in the world. However, lithium mining often leads to deforestation and water pollution. The lithium battery production process also generates high carbon emissions which require fossil energy which require fossil energy. On the other hand, Indonesia, as the largest corn producer with 25.3 million tons of corn by 2024, generates a significant amount of corn waste. This waste can be turned to produce carbon dots (C-Dots), which, when synthesized with AgNO_3 , serve as anode materials for batteries. AgNO_3 can be used as anode material in batteries. This study compares four types of C-Dots from corn waste: husk, stalk, leaves, and cobs. From the FTIR results, analysis of C-Dots derived from corn waste reveals significant changes in functional groups before and after treatment with AgNO_3 , which is crucial for their application as battery anodes. Potential testing voltage and electric current were also tested using multimeter and swagelok. The measurement results showed that corn husk C-Dots had a voltage of 51.6 mV and a current of 8 mA, C-Dots corn stalks have a voltage of 5.4 mV and a current of 5.2 mA, corn cob C-Dots of 11.4 mV and a current of 3.8 mA, and corn leaf C-Dots had a voltage of 149.6 mV and a current of 2.9 mA. Based on the multimeter test conducted, it was found that corn leaf C-Dots had the best potential as an anode battery.

Keywords: Lithium Battery, C-Dots, Corn Waste

1. Introduction

1.1 Background

Indonesia is the 8th largest corn producing country in the world by 2023. Corn waste in Indonesia continues to increase along with the growth of corn production which is projected to reach 25.3 million tons in 2024. Each harvest produces waste in the form of corn husk, stalks, leaves, and cob which can amount to more than 30% of the total plant weight. Most of this waste is usually burned directly, leading to air pollution and carbon emissions. However, this waste has great potential to be turned into biomass, a renewable energy source that can be used in power plants, helping to reduce coal use and negative environmental impacts.

At the same time, Lithium-ion batteries pose a serious risk to the environment if not managed properly. They contain toxic heavy metals such as

lithium, cobalt, nickel and manganese, which can contaminate soil and water when batteries leak or are disposed of carelessly. Exposure to these heavy metals can negatively affect ecosystems and human health through the food chain. In addition, lithium-ion battery waste also has the potential to cause fires or explosions due to the flammable electrolyte material, especially if the battery is damaged or exposed to heat. The recycling process of these batteries is difficult and often inadequately available, so used batteries often end up in landfills, which increases the risk of environmental pollution.

Therefore, this research was carried out which can reduce the adverse effects of using waste batteries while utilizing corn waste that is wasted. This paper explores the synthesis of these carbon dots, their properties, and their potential as a sustainable and environmentally friendly alternative to lithium-ion batteries. The research presented in this

paper has the potential to contribute to the development of more sustainable and environmentally friendly technologies for the future.

1.2 Objection

1. Does corn waste can be used as an alternative material in the manufacture of environmentally friendly battery?
2. How is the effectiveness comparison between C-dots from Corn Husk, Corn stalks, Corn Leaves, and Corn Cob as an environmentally friendly battery?
3. How do the FTIR results on C-Dots and C-Dots/Ag compare?

1.3 Purpose

1. To identify the potential of corn waste as an alternative raw material in the manufacture of environmentally friendly batteries.
2. Testing the effectiveness of C-dots from Corn Husk, Corn Stalks, Corn Leaves, and Corn Cob as environmentally friendly batteries.
3. Analyzed the effectiveness of C-Dots and C-Dots-Ag based on many forms of its functional groups.

1.4 Aim

1. For the environment, providing an alternative utilization of corn waste and also prevents the adverse effects of using hazardous waste in the form of batteries.
2. For health, reducing exposure to hazardous chemicals from waste lithium batteries that cause respiratory problems, heart disease, and cancer, using carbon-based technology from corn waste.

1.5 Basic Theory

1.5.1 Carbon Nanodots

Carbon dots (C-dots) are new types of fluorescent carbon nanomaterials and are among the most significant members within the carbon nanomaterials family. Generally, C-dots are finely dispersed spherical particles with a size smaller than 10 nm (Kang C., etc. 2020). The

controllable production of carbon dots is currently in its initial stages of development. Synthesis methods can be categorized into two main types: top-down and bottom-up approaches. Bottom-up methods include techniques like hydrothermal, microwave, thermal decomposition, carbonization, pyrolysis, solvothermal, and ultrasonic. Top-down methods include laser ablation, arc discharge, and electrochemical (Khayal A., etc. 2021). The raw materials for creating carbon dots are plentiful and can be categorized into two types: organic and inorganic carbon sources. In practical applications, organic carbon sources such as organic compounds, organic products, and biomass waste are more commonly utilized for synthesizing carbon dots (Cui L., etc. 2021). The use of different synthesis methods and different base materials will result in differences in the levels of carbon, oxygen, and nitrogen formed, surface structure, and functional groups resulting in different characteristics.

1.5.2 Corn Waste

Corn, which also known as *Zea mays L.* belongs to the *Gramineae* family or grasses and is classified as a Monocotyledonous plant. The morphological structure of the corn plant consists of roots, stems, flowers, and fruit (Syukur M.). There are several local Indonesian/regional terms for various kinds of corn plant waste or by-products of corn-based industries such as corn stubble (the entire corn plant including stems, leaves and young corn kernels), corn straw (parts of corn stems and leaves that have been left to dry in the field and harvested when the corn cob is picked), corn husk (the outer skin of corn kernels which is usually discarded), corn cob (waste obtained when corn kernels are threshed from the fruit) (Bunyamin Z., etc. 2014).

1.5.3 AgNO₃

AgNO₃ or silver nitrate compounds are metal compounds derived from silver

metal (Ag) reacted with concentrated nitric acid solution (HNO₃ 8 M) which reacts according to the following reaction equation $6\text{Ag (s)} + 8\text{HNO}_3 \text{ (aq)} \rightarrow 6\text{AgNO}_3 \text{ (aq)} + 4\text{H}_2\text{O (l)} + 2\text{NO (g)}$ (Addiin I., etc. 2021). This corrosive compound is arranged in a planar trigonal array and has a molecular weight of 169.873g/mol, Exact Mass of 168.89291 g/mol, Monoisotopic Mass of 168.89291 g/mol, Heavy Atom Count of 5, and Complexity of 18.8 (PubChem. 2021). Silver nitrate remains stable under light exposure and easily dissolves in water, breaking down into its component ions. It finds use in diverse fields such as photography, the creation of silver-containing explosives, and the manufacture of certain medical substances

assembly and disassembly in an airtight system. Swagelok type cell is specially designed for lithium-ion battery's research. Cells are useful for studying the behavior of single or full electrodes using foil or carbon cloth electrodes as anode and cathode electrodes, this is to achieve super capacitors or Li-ion batteries that are close to the device system with very minimal steps and can perform various types of measurements such as CV, EIS, and Charge-discharge.

1.5.4 Lithium-Ion Battery

Lithium is one of the minerals that has the highest demand in the fourth Industrial Revolution. The main raw material source for batteries is Lithium. (Salafudin. 2021) Lithium is an irreplaceable component in Lithium-ion batteries, and is used as a cathode, electrolyte and anode. Lithium is preferred as a battery material because it is the lightest metal with the highest electrochemical potential among all metals, and has the highest level of energy density among all metals (Bae H., etc. 2021).

1.5.5 Swagelok

The swagelok type cell battery fixture is used for lithium batteries, Zinc ion batteries, sodium ion batteries and solid battery research work, it can replace of coin cell cases to save the cost. A swagelok is a simple cylindrical electrochemical cell consisting of two electrodes, a separator, and a Swagelok fitting, which is a stainless-steel compression fitting that holds the cell assembly together and provides an electrical connection to the electrodes. The Swagelok cell is a custom electrochemical cell that allows for easy

2. Method and Experimental Details

2.1 Time and Place

This research was conducted in July 2024 at researcher's house and STIFAR Laboratory.

Date	Activity	Activity Details	Place
30 June – 10 July 2024	Planning	Literature studies about Corn, Carbon Nanodots, Battery cells, AgNO ₃ , Swagelok, and FTIR Method	Researcher's house
11 – 13 July 2024	Research Design	Looking for tools and material for research proses	Lab store & Corn fields
14 – 21 July 2024	Research Process	Making C-dots, FTIR Test, AgNO ₃ synthesis, Making AgNO ₃ C-dots Battery Tablet	Researcher's house & STIFAR Laboratory
22 – 31 July 2024	Analyze	Discussion to analyze and write the research result	Researcher's house

2.2 Research Design



2.3 Experiment Method

2.3.1 Tools and Material

Material needed for research are:

1. Dry Corn Husk
2. Dry Corn Stalks
3. Dry Corn Leaves
4. Dry Corn Cob
5. Aquadest
6. AgNO_3 crystals
7. KOH
8. Ethylene Glycol
9. 70% Alcohol
10. Paper filters

Tools needed for research are:

1. Microwave
2. Beaker glass
3. Watch glass
4. Porcelain mortars
5. Mg/Micro unit scales
6. Swagelok
7. Multimeter
8. Laboratory water bath
9. Blender
10. Knife

2.3.2 Experiment Process

2.3.2.1 Making Carbon Dots

1. Prepare all tools and material
2. Cut material into small pieces



3. Blend with blender to create smaller particle



4. Microwave at 450 watts for 20 minutes x 5



5. Filter Carbon Dots with aquadest



6. Dry Carbon Dots for 6 hour



7. Grind Carbon Dots into fine particles using mortar



8. Carbon dots are ready to be used

2.3.2.2 Synthesis of C-dots with AgNO_3

1. 10g of AgNO_3 is added into 10g of C-dots that has been dispersed into 1000ml of 70% alcohol.



2. 300ml of ethylene glycol and 2g of KOH is also added. Then, reduced in 85°C water bath while being stirred constantly for 60 minutes.



3. Separate the sediment from the solution.
4. Wash the sediment with aquadest and dried at 50°C overnight in an oven.
5. Reduced C-dots without AgNO_3 is also prepared as comparison

2.3.2.2 Making & Testing CDs-Maize Power Tablet

1. Insert 1g of C-dots into Swagelok (pressing tools)

- The process of pressing is done manually until a tablet is formed.
- Reduced C-dots without AgNO_3 is also prepared for independent study as a comparison.
- Test and compare the result CDs-Cornergy millivolt and microampere using multi meter.

3. Result and Discussion

3.1 Multi-meter Test

a. C-dots

No	C-dots	mV	mA
1	Corn Husk	1.4	0
2	Corn Stalks	0.3	2
3	Corn Leaves	10.6	0
4	Corn Cob	0.4	0

b. C-dots/Ag

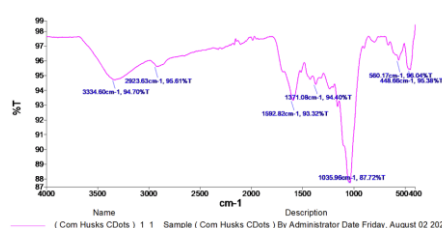
No	C-dots/Ag	mV	mA
1	Corn Husk	51.6	8
2	Corn Stalks	5.4	5.2
3	Corn Leaves	149.6	2.9
4	Corn Cob	11.4	3.8

Based on the result of Multimeter test on 4 C-dots/Ag which are Corn Husk, Corn Stalk, Corn Leave, and Corn Cob, it can be seen that Corn Leave C-dots produces the highest voltage with 149.6mV and 2.9mA compared to Corn Husk C-dots/Ag with 51.6 mV and 8mA, Corn Stalks C-dots/Ag with 5.4mV and 5.2mA, and Corn Cob C-dots/Ag with 11.4mV and 3.8mA. It can be concluded that we can utilize C-dots/Ag from Corn Leave as an ingredient to construct an alternative to lithium-ion batteries.

3.2 FTIR Test

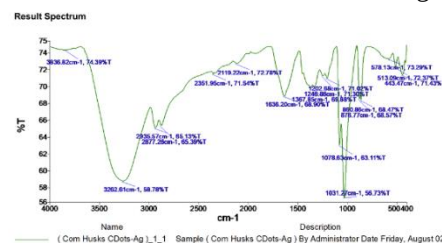
a. Corn Husk C-dots

FTIR Result of Corn Husk C-dots



FTIR Vibration modes	Wavenumber (cm^{-1})	Transmittance (%T)
OH	3334.60	94.70
C-H	2923.63	95.61
N=H	1592.82	93.32
C=C	1371.08	94.40
C-C	1035.95	87.72

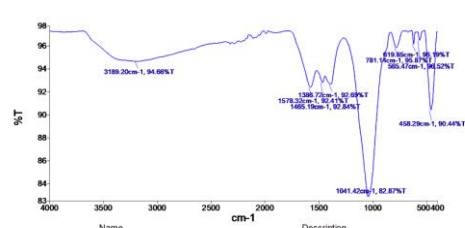
FTIR Result of Corn Husk C-dots/Ag



FTIR Vibration modes	Wavenumber (cm^{-1})	Transmittance (%T)
OH	3262.61	58.78
C-H	2877.26	65.39
N=H	1636.20	68.9
C=C	1246.86	71.3
C-C	1031.27	56.73

b. Corn Stalks C-dots

FTIR Result of Corn Stalks C-dots



FTIR Vibration modes	Wavenumber (cm^{-1})	Transmittance (%T)
OH	3189.20	94.66
N=H	1386.72	92.69
C=C	1465.19	92.84
C-C	1041.42	82.87

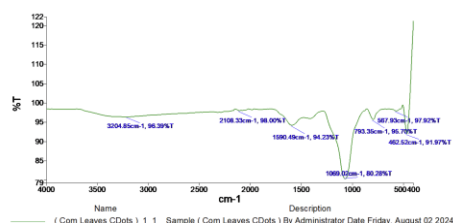
FTIR Result of Corn Stalks C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3263.19	39,85
C-H	2879.00	48,75
N=H	1636.59	53,01
C=C	1332.91	54,16
C-C	1031.77	39,05

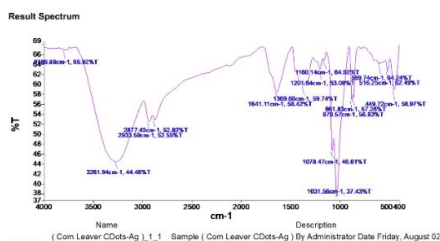
c. Corn Leaves C-dots

FTIR Result of Corn Leaves C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
N=H	1590.49	94.23
C-C	1069.02	80.28

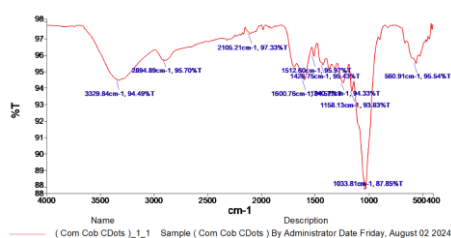
FTIR Result of Leave C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3261.94	44,46
C-H	2933.59	52,55
N=H	1641.11	58,42
C=C	1369.66	59,74
C-C	1031.56	37,43

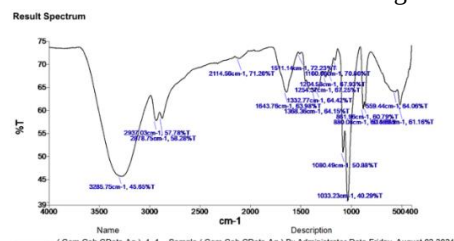
d. Corn Cob C-dots

FTIR Result of Corn Cob C-dots



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3329.84	94.49
C-H	2894.89	95.70
N=H	1600.76	94.52
C=C	1426.75	95.43
C-C	1033.81	87.85

FTIR Result of Corn Cob C-dots/Ag



FTIR Vibration modes	Wavenumber (cm ⁻¹)	Transmittance (%T)
OH	3285.75	45,65
C-H	2937.03	57,78
N=H	1643.76	63,98
C=C	1332.77	64,42
C-C	1033.23	40,29

The results of the FTIR testing of C-dots are presented in the figure and table above. Based on the FTIR analysis of C-dots before and after the AgNO₃ synthesis treatment, differences in functional group amount were observed. In Corn Husk C-dots, the OH bond was detected at a wavenumber of 3334.60 cm⁻¹ with a transmittance of 94.70%, while the C-H bond was observed at a wavenumber of 2923.63 cm⁻¹ with a transmittance of 95.61%. Conversely, in Corn Husk C-dots/Ag, the OH bond was identified at a wavenumber of 3262.61 cm⁻¹ with a transmittance of 58.78%, and the C-H bond was detected at a wavenumber of 2877.26 cm⁻¹ with a transmittance of 65.39%.

This also applies to other C-dots. In Corn Stalks C-dots, the OH bond was detected at a wavelength of 3189.20 cm⁻¹ with a transmittance of 94.66%, whereas the C-H bond was not detected. In Corn Stalks C-dots/Ag, the OH bond was identified at a wavelength of 3263.19 cm⁻¹ with a transmittance of 39.85%, while the C-H bond was also observed.

The FTIR results for Corn Leaves C-dots indicated that neither the OH nor C-H bonds were detected. However, in Corn Leaves C-dots/Ag, the OH bond was detected at a wavelength of 3261.94 cm⁻¹ with a transmittance of 44.46%, and the C-H bond was observed at a wavelength of 2933.59 cm⁻¹ with a transmittance of 52.55%.

FTIR results for Corn Cob C-dots, OH bond was detected at a wavelength of 3329.84 cm^{-1} with a transmittance of 94.49%, and the C-H bond was detected at a wavelength of 2894.89 cm^{-1} with a transmittance of 95.70%. On the other hand, in Corn Cob C-dots/Ag, the OH bond was identified at a wavelength of 3285.75 cm^{-1} with a transmittance of 45.65%, as well as the C-H bond at a wavelength of 2937.03 cm^{-1} with a transmittance of 57.78%.

C-Dots, which are carbon nanoparticles less than 10 nm in size, have good electrical conductivity and chemical stability properties, therefore serving as an efficient anode for energy storage. When the system is charged, silver ions from AgNO_3 dissociate and interact with C-Dots, allowing silver ions to accumulate on the surface or in the C-Dots matrix, creating an active area that stores charge. During charging, silver ions participate in redox reactions, experiencing reduction and attaching to C-Dots, which increases the charge storage capacity. When used, the accumulated silver ions are released back into solution, generating electricity. C-Dots also aid in the even distribution of ions and reduce internal resistance, while AgNO_3 improves ionic conductivity and ion transfer efficiency.

It is seen from the FTIR results, analysis of C-Dots derived from corn waste reveals significant changes in functional groups before and after treatment with AgNO_3 , which is crucial for their application as battery anodes. For C-Dots sourced from corn husks, there was a notable decrease in transmittance for hydroxyl (OH) groups from 94.70% to 58.78% and for C-H groups from 95.61% to 65.39%. This reduction suggests a reaction between silver ions and hydroxyl groups, potentially enhancing ionic conductivity and the stability of the electrode. Similarly, C-Dots from stalks and cobs exhibited an increase in C-H groups post-treatment, indicating improved energy storage potential

through enhanced redox reactions, which is essential for boosting battery charge and discharge efficiency.

On the opposite, C-Dots derived from leaves, which initially showed no OH or C-H groups, successfully produced both groups following AgNO_3 treatment, albeit with relatively low transmittance. This indicates that the synthesis process not only introduced new functional groups but also increased the material's suitability for battery applications. Overall, the detectable hydroxyl groups and the added C-H groups in most samples suggest that C-Dots from corn waste, after modification, have significant potential as battery anodes due to their favorable electrochemical properties for efficient energy storage.

4. Conclusion

From the research that has been done, it can be concluded that C-Dots/Ag from corn waste has the best potential as a material for making alternative lithium-ion battery.

- a. C-Dots sourced from corn waste and combined with C-Dots sourced from corn waste and combined with AgNO_3 have the potential to create a sustainable and environmentally friendly lithium-ion battery alternative.
- b. C-Dots/Ag made from corn leaves has the best potential to be an alternative to lithium-ion batteries with the highest voltage of 149.6 mV and a current of 2.9 mA.
- c. FTIR results show that the synthesis of AgNO_3 on C-Dots from corn waste causes significant functional group changes. corn waste caused a significant change in functional groups, which correlated with an increase in voltage of correlated with an increase in voltage of 139.0 mV and current of 2.9 mA on corn leaves.

5. Acknowledgements

Finally, our project is finished. We would like to express our deepest gratitude to Drs. Agus Priyatno, M.Pd., for his invaluable guidance, support, and encouragement throughout the course of this research. His expertise and insights have been instrumental in the successful completion of this study. We also extend our appreciation to all those who have contributed to this project, directly or indirectly, for their assistance and cooperation.

6. References

- [1] Addiin I., Yamtinah S. 2021. Pembuatan Perak Nitrat (AgNO₃) Teknis dari Limbah Penyepuhan Perak. *Seminar Nasional Pendidikan Sains VI 2016*, Surakarta: Universitas Sebelas Maret; <https://www.neliti.com/id/publications/173466/pembuatan-perak-nitrat-agno3-teknis-dari-limbah-penyepuhan-perak#cite>
- [2] Bae, H., & Kim, Y. (2021). Technologies of lithium recycling from waste lithium ion batteries: a review. *Materials advances*, 2(10), 3234-3250. <https://pubs.rsc.org/en/content/articlehtml/2021/ma/d1ma00216c>
- [3] Bunyamin, Z., Efendi, R., Andayani, N. N., & Serealia, T. (2013). Pemanfaatan limbah jagung untuk industri pakan ternak. In *Seminar Nasional Inovasi Teknologi Pertanian* (Vol. 153)
- [4] Cui L., Ren X., Sun M., Liu H., Xia L. (2021). Carbon Dots: Synthesis, Properties and Applications. *Nanomaterials*, 11(12), 3419. doi:10.3390/nano11123419; <https://www.mdpi.com/2079-4991/11/12/3419>
- [5] Hasibuan, R. (2016). Analisis dampak limbah/sampah rumah tangga terhadap pencemaran lingkungan hidup. *Jurnal Ilmiah Advokasi*, 4(1), 42-52.
- [6] Kang, C., Huang, Y., Yang, H., Yan, X. F., & Chen, Z. P. (2020). A review of carbon dots produced from biomass wastes. *Nanomaterials*, 10(11), 2316. doi:10.3390/nano10112316; <https://www.mdpi.com/journal/nanomaterials>
- [7] Khayal, A., Dawane, V., Amin, M. A., Tirth, V., Yadav, V. K., Algahtani, A., & Jeon, B. H. (2021). Advances in the methods for the synthesis of carbon dots and their emerging applications. *Polymers*, 13(18), 3190. doi:10.3390/polym13183190; <https://www.mdpi.com/journal/polymers>
- [8] PubChem. 2024. Bethesda (MD): National Library of Medicine (US), National Center for Biotechnology Information. <https://pubchem.ncbi.nlm.nih.gov/compound/25199729> [2024 July 5].
- [9] Salafudin, S. (2021). Sumberdaya alam lithium Indonesia. *Rekayasa Hijau: Jurnal Teknologi Ramah Lingkungan*, 5(2), 178-187. <https://ejurnal.itenas.ac.id/index.php/rekayasahijau/article/view/4385>
- [10] Syukur M., Rifianto A. (2014). Jagung Manis. 8-10. Penebar Sawdaya, Cibubur, Jakarta.